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Cartilaginous microtissues exhibit extreme resilience under compression with size-dependent mechanical properties

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Abstract

Self-assembled cartilaginous microtissues provide a promising means of repairing challenging skeletal defects and connective tissues. However, despite their considerable promise in tissue engineering, the mechanical response of these engineered microtissues is not well understood. Here we examine the mechanical and viscoelastic response of progenitor cell aggregates formed from human primary periosteal cells and the resulting cartilaginous microtissues under large deformations as might be encountered *in vivo*. We find that the mechanical response of these tissues is strongly size dependent due to surface tension effects, with a scaling law for the Young's modulus of $E \propto D^m$, where D is the diameter of the tissues, and m varies with the tissue type. Similar size effects are found to govern the interfacial surface tension and the viscosity. In addition, these microtissues are extremely resilient, as they sustain over 90 % of compressive strain without mechanical failure. Stress relaxation experiments reveal a fast stress dissipation at short time scale within a few seconds, followed by oscillations in measured stresses that depend on actomyosin contractility. In summary, these experiments reveal the remarkable and unanticipated resilience of cartilaginous microtissues under large mechanical strains, a property that may facilitate their use in a variety of tissue engineering applications. More broadly, our data highlight the importance of surface tensions in determining the mechanical properties of tissues on the micron and the mm length scales.

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Declaration of competing interest

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CRediT authorship contribution statement

Charalampos Androulidakis: Writing – original draft, Methodology, Investigation, Formal analysis, Conceptualization. **Hanna Svitina:** Visualization, Methodology, Investigation. **Konstantinos Ioannidis:** Visualization, Methodology, Investigation. **Alexander R. Dunn:** Writing – review & editing, Validation, Data curation. **Ioannis Papantoniou:** Writing – original draft, Validation, Supervision, Resources, Data curation, Conceptualization.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.biomaterials.2024.123074.

1. Introduction

Self-assembled, three-dimensional (3D) cell spheroids constitute a promising class of models for tissue regeneration, drug screening, and disease modeling due to the facile manipulation of their size and composition [1]. In particular, cell spheroids assembled from cartilage precursors show promise in the repair of damage and degeneration in bone and cartilage [2–4]. Importantly, individual cartilage spheroids must possess robust mechanical integrity to avoid mechanically induced failure *in vivo*. More broadly, the solid tissues that make up the human body must withstand appreciable mechanical loads as part of their physiological functions [5]. Understanding how living tissues respond to large, external loads is thus a critical aspect of the design and development of cell-based regenerative medicine therapies.

Biological tissues exhibit a diversity of mechanical behaviors such as nonlinear viscoelastic response and strain stiffening under large deformations [5]. To date, the mechanical properties of spheroids with diameters in the micron scale have been tested under uniaxial compression using parallel plate squeezing experiments [6–12] and micropipette aspiration under pressure [13–15], while cell monolayers in the form of sheets have been tested under tension [16–18]. As multicellular structures, the mechanical properties of tissues depend on the mechanics of their single cell components, and therefore stiffer single cells results in stiffer cell aggregates and tissues [19]. The presence of extracellular matrix (ECM) also plays a crucial role in the overall mechanical properties resulting in stiffening of the tissue compared to cell aggregates [12,19]. Furthermore, cell-cell and cell-ECM junctions lead to intercellular force transmission that in turn affects the overall tissue mechanics [20]. In addition to elasticity, living tissues are also viscoelastic, meaning that their mechanical properties are time dependent and relax over time under constant strain to dissipate mechanical stresses. Stress relaxation measurements have been reported for various types of cell sheets and spheroids [19,21]. Surface tension also plays a pivotal role in the mechanics of biological matter, from single cell to tissue levels [12]. Within cell aggregates, inhomogeneities in mechanical stress [22] have been interpreted as reflecting the presence of a softer core surrounded by a stiffer ring-like structure characterized by increased cell surface tensions and the presence of multicellular actin cables [23–25]. In these studies [23–25], spheroids made from various cell lines were tested including colon carcinoma cells (CT26), fibroblasts isolated from human (HS-5), and fibroblasts 3T3 derived from mice and rats.

The mechanical properties of cell spheroids broadly, and cartilaginous microtissues in particular, remain incompletely understood. Microtissues used for tissue engineered implants destined for skeletal defect regeneration will be exposed to mechanical stresses during and post implantation. This applies for both long bone defects as well as osteochondral defects of the knee joint. In both cases a transition from the implanted immature cartilage to hypertrophic cartilage leading to mineralization and bone formation will be processes that these microtissues will need to undergo. Hence, understanding how engineered cartilaginous microtissues respond to mechanical deformation at progressive degrees of chondrogenic maturation will be critical in defining the ideal microtissue-based implant properties. In this work we strived to assess this also taking into account high loads in order to explore mechanical regimes that have not been to date systematically explored for cartilaginous microtissues in the context of skeletal tissue engineering. Prior work exploring mechanical properties of *in vitro* engineered tissues, and their compressive response has been limited to maximum compressions of 50 % [6–12]. However, the behavior of tissues under large deformations is potentially important, as excessive mechanical loading can cause damage to the tissues, and to the fate of implanted tissue engineered constructs in defects where high loads are generated such as osteochondral defects of the knee joints [26,27]. In a recent study [28], compressive strains of 80–85 % were required to induce Ca^{2+} signaling in chondrocytes from articular cartilage of pigs, showing that the signaling via piezo channels regulates biomechanical responses under large deformations. This result suggests that signaling pathways related to inflammation, disease triggering and cell death might occur under supraphysiological levels of mechanical compressive strains which can be higher than 80 %. While single cell chondrocytes sustained such high compressive mechanical loads [28], it is not known if *in vitro* cartilaginous microtissues possess similar resilience and how their mechanical properties are altered under high compression.

Another issue concerning the use of cartilaginous microtissues for tissue engineering applications is related to tissue scalability. Scaffold free strategies advocate the use of microtissues as building blocks [29–31] with microtissue dimensions ranging from ~100 to 400 μm in diameter, while implant dimensions can reach up to the cm scale as in the case of long bone or osteochondral defects [2] requiring the development of larger engineered and load bearing tissues. Recent results have shown significant differences in the viscoelastic properties across tissue length scales, for example in comparing single cancerous cells and tumors [32], and a length scale dependence for stress transmission within tumors in *in vivo* mouse models [33]. Understanding how the mechanical properties of engineered tissues vary across scales and in response to large deformations is thus of fundamental importance for designing improved tissue engineered products, but these topics remain largely unexplored.

Human periosteum derived cells (hPDCs) are the main contributors to the process of endochondral ossification during fracture healing [34] and have been used for the engineering of cartilaginous microtissues able to regenerate critical-size long bone defects [2]. Besides hPDCs additional cell types such as bone marrow mesenchymal stem cells (MSCs [35]) and adipose-derived MSCs [36] can also be used to derive functional cartilaginous microtissue structures. The response of such cartilaginous microtissues to large physical deformations comparable to those that might occur *in vivo* is, however, currently unknown.

In this work we exposed hPDC cell aggregates and cartilaginous microtissues of increasing chondrogenic maturity to compressive strains that exceeded 90 %. We found that the effective stiffness of these microtissues was strongly dependent on size. This effect is attributed to surface tension for the undifferentiated cell aggregates, while cartilaginous microtissues demonstrated progressive increases in stiffness during maturation due to the secretion of collagen fibers. In both cases, the Young's modulus (E) of the tissues scaled as $E \propto D^m$, where D is the diameter of the tissues and m varies with the tissue type. In addition, microtissues showed strain stiffening under large deformations, and sustained compressions of over 90 % without mechanical failure. Microtissues also possessed viscoelastic properties, and stress relaxation experiments reveal that they dissipate stress rapidly following a power law. The initially abrupt stress relaxation is followed by a highly oscillatory response with stress fluctuations. This response deviates from the viscoelastic solid or viscoelastic fluid behavior and presents another mechanism of higher complexity with possible implications in biological function. The present results provide insights in the mechanical response of *in vitro* tissues under large compressive deformations, which are relevant to their effective use for skeletal tissue engineering.

2. Experimental

Cell Expansion:

hPDC cells were cultured up to passage 9 in growth medium consisted of DMEM with high glucose (Life Technologies, UK) supplemented with 10 % fetal bovine serum (HyClone FBS, Thermo Scientific, USA), 1 % antibiotic–antimycotic (100 units/mL penicillin, 100 mg/mL streptomycin, and 0.25 mg/mL amphotericin B), and 1×10^{-3} M sodium pyruvate (Life Technologies, UK) at 37 °C, 5 % CO_2, and 95 % humidity. The growth medium was refreshed two to three times weekly until the cells reached 90 % confluency, at which point they were collected using TrypLE™ Express (Life Technologies, UK). All experimental procedures were ethically approved by the Human Medical Research Ethical Committee at Katholieke Universiteit Leuven, and informed consent was obtained from all patients (ML7861). For the chondrogenic differentiation, a chemically defined chondrogenic media (CM) previously established and validated in our lab [2], which consists of LG-DMEM (Gibco) supplemented with 1 % antibiotic-antimycotic (100 units/mL penicillin, 0.25 mg/mL amphotericin B and 100 mg/mL streptomycin), 40 μg/mL proline, 1 mm ascorbate-2 phosphate, 100 nM dexamethasone, ITS + Premix Universal Culture Supplement (Corning) (including 6.25 μg/mL insulin, 6.25 μg/mL transferrin, 6.25 μg/mL selenious acid, 1.25 μg/mL bovine serum albumin (BSA), and 5.35 μg/mL linoleic acid), 20 μm of Rho-kinase inhibitor Y27632 (Axon Medchem), 100 ng/mL GDF5 (PeproTech), 100 ng/mL BMP-2 (INDUCTOS®), 10 ng/mL TGFβ1 (PeproTech), 1 ng/mL BMP-6 (PeproTech) and 0.2 ng/mL FGF-2 (R&D systems).

Formation of cell aggregates and cartilaginous microtissues:

For the formation of microtissues we used low attachment well-plates. Seeding densities starting from 5k cells until 100k with increment of 5k were used to produce tissues of varying dimensions. A detailed quantification of the seeded number of cells and the resulting dimensions is given in the SI. Cell aggregates were formed by collecting and seeding the cells at varying concentrations, which then self-assembled into microtissues. For the cartilaginous microtissues, hPDCs were initially collected in a tube, centrifuged, and the expansion medium was removed. Next, the cells were resuspended in chondrogenic medium used in previous studies [2,37] and seeded at various concentrations in low adherent well-plates, where they differentiated towards a cartilage phenotype over the course of 21 days [2]. The chondrogenic medium is a serum-free and chemically defined solution designed for chondrogenesis. In order to confirm the chondrogenic differentiation of the tissues used herein, we examined by immunofluorescent staining the marker Osterix (OSX) and the proliferation marker Ki-67, as seen in Fig. S3. The OSX is markedly increased from day 1 to day 7 (Fig. S3) and along with the collagen secretion shows the differentiation of the cell aggregates to cartilaginous phenotype. The proliferation decreases as the staining for Ki-67 shows (Fig. S1), in agreement with the reduction observed in the size of the microstissues. In order to confirm efficient chondrogenic differentiation, we performed histological sectioning and immunostaining indicating the progressive secretion of extracellular matrix which was positive for glycosaminoglycans (GAG) as evidenced through Alcian Blue stain (Fig. S2) as well as the presence of sulfated GAGs as evidenced by Safranin O staining (as presented below in Fig. 4). In contrast cell aggregates did not show any positivity for this stain.

For these microtissues the seeded number of cells corresponds to 5k. In Fig. S1, the same measurements are presented but this time for larger microtissues produced by seeding 50k cells, in order to examine structural differences between microtissue size. Moreover, the ECM compartment of the microtissues, stained positively for collagen type I and II (Fig. S2), for both microtissue sizes during chondrogenic differentiation. The results (Fig. S2) show the presence of both collagen types in the secreted ECM and further confirm the cartilaginous phenotype of these microtissues across dimensions examined in this work.

Mechanical testing:

mechanical experiments performed using a MicroTester LT (CellScale) device. Cell aggregates and cartilaginous microtissues were tested live in a phosphate buffered saline (PBS) bath at 37° C. The samples were deposited on a steel substrate inside the PBS bath without fixation since they remained immobilized during compression and no slippage occurred during the experiment as deduced from the side camera which recorded the whole experiment. Strain-controlled compression experiments were performed using tungsten beams with stiffness 4.11 GPa and diameters of 0.40 and 0.55 mm to apply mechanical loads and record force-displacement curves, with force resolution of 9.658 μN and 34.52 μN, respectively. The maximum applied forces were ~3000 μN for the cell aggregates and ~6000 μN for the cartilaginous microtissues, respectively. We note that this device has certain limitations when high forces are applied. The thicker beam can reach higher forces, but the force resolution is compromised. Thus, the measurements kept in the range that the forces can be reliably recorded and simultaneously provide sufficient force resolution. A side camera with a 2× zoom objective was used to record the deformation, force and measurement of the size of the spheroids with sub-micron scale resolution. All experiments performed with a constant strain rate at 2 % sec^{-1} which falls within the recommended rate for testing cartilage under compression [38], and with a maximum applied compressive strain of 95 %. Stress relaxation experiments were performed by compressing the tissues to strains of 30 % and 50 % and recording the force variation over a minute time scale. Analysis performed using the MicroTester software, which allows the direct estimation of interfacial surface tension at a given level of deformation according to previous theoretical approach [11,39]. The viscosity of the tissues is estimated using the Maxwell model. The part of the stress relaxation response from the peak load until the first oscillation was fitted with an equation in the form of $\sigma(t) = \sigma_0 exp(-t/\tau)$, where σ is the stress, t is the time and τ is the relaxation time. The viscosity calculated by $\eta = E\tau$, where E is the Young’s modulus and τ is the relaxation time obtained by the stress relaxation experiments.

Whole-microtissue immuno-staining:

Microtissues were fixed in 4 % PFA at 4 °C for 1 h and washed thoroughly with PBS. 5–10 microtissues per group were transferred into 1 well of the U-bottom Ultra-low Attachment 96 well plate (Corning). Microtissues were stained with 1× Alexa Fluor™ 488 phalloidin (A-12379, Invitrogen) in 0.2 % Triton X-100 in PSB for 2 days on shaker at 4 °C. Next, phalloidin solution was removed and the microtissues were incubated with anti-human Osterix/Sp7 monoclonal antibody (MAB7547, R&D Systems) or 1:200 mouse anti-Vimentin (Clone V9) (NBP1–97670, Novus Biological) in PBS with 2.5 % BSA and 0.2 % triton-X100 for 2 days on shaker at 4 °C. Microtissues were washed 5 times for 1 h in PBS on shaker at room temperature. Secondary antibodies were diluted in the same buffer as primary antibodies: 1:500 Goat Anti-Rabbit IgG H&L (Alexa Fluor® 594) (ab150080, Abcam) and incubated for 2 days on shaker at 4 °C. Nuclei were counterstained with 1:2000 SYTOX Deep Red Nucleic Acid Stain (S11380, Invitrogen) for 2 h at room temperature. Then, microtissues were washed in PSB overnight and cleared and mounted using RapiClear 1.52 (SunJin Lab) and iSpacer (SunJin Lab) to control the height of chamber. Confocal and two-photon imaging was performed using a Zeiss LSM 780 microscope.

Histological Staining:

Microtissues from different groups were fixed at 4 °C in freshly prepared 4 % paraformaldehyde overnight. The samples were then dehydrated through a series of ethanol gradients and embedded in paraffin. Paraffin blocks were sectioned at 5 μm thickness using a microtome. The sections were deparaffinized twice in 100 % HistoClear for 5 min each and rehydrated through a decreasing ethanol gradient (96 %, 70 %, and 50 %). After rehydration, the sections were stained with either 0.2 % (w/v) Safranin O to assess sulfated glycosaminoglycans (sGAG) or 1 % (w/v) alcian blue 8GX in 0.1 M HCl to assess total glycosaminoglycans (GAGs). Stained sections were visualized using an Olympus IX83 inverted microscope (Olympus, Tokyo, Japan) equipped with a DP73 camera. Raw images were processed and analyzed using QuPath software (version 0.5.0).

Immunohistochemistry:

Microtissue sections from different groups were deparaffinized twice in 100 % HistoClear for 5 min each and rehydrated through a decreasing ethanol gradient (96 %, 70 %, and 50 %). Antigen retrieval was performed by enzymatic treatment with 0.1 mg/mL pepsin in 0.2 M HCl for 15 min at room temperature. To minimize nonspecific reactivity, sections were incubated in blocking buffer (PBS with 2.5 % BSA) at room temperature for 30 min. Primary antibodies were diluted in PBS containing 2.5 % BSA and 0.3 % Triton X-100. Tissue sections were incubated overnight at 4 °C with the following antibodies: 1:200 mouse anti-human COL1A (COL-1) antibody (sc-59772, Santa Cruz Biotechnology) and 1:50 rabbit anti-human collagen type II polyclonal antibody (AB761, Merck). After three 10-min washes in PBS, the sections were incubated overnight at 4 °C with the following secondary antibodies, diluted in the same buffer as the primary antibodies: 1:500 goat anti-mouse IgG (H + L), Alexa Fluor™ 488, and 1:500 donkey anti-rabbit IgG (H + L) highly cross-adsorbed secondary antibody, Alexa Fluor™ Plus 594 (A-32754, Invitrogen). Nuclear counterstaining was performed using 1 μg/mL DAPI in PBS for 10 min at room temperature. Slides were mounted with Mowiol 4–88 (Sigma-Aldrich, Merck KGaA) and visualized using an Olympus IX83 inverted microscope (Olympus, Tokyo, Japan) equipped with a DP73 camera.

Confocal and 2-photon imaging:

Confocal and two-photon z-stacks were recorded on an LSM 780 microscope (Zeiss) using a 25×, water-immersion objective (NA 0.8, Zeiss). For confocal z-stacks Alexa-594, SYTOX Deep Red Nucleic Acid Stain were excited by 561 or 633 respectively. Emission signals were descanned and collected with spectral detectors. A tuneable Mai Tai DeepSee Titanium-Sapphire femtosecond laser (680–1050 nm; Spectra-Physics) was used for recording 2-photon z-stacks of forward second harmonic generation, Alexa-488 with an excitation wavelength set at 850 nm. Images were corrected for brightness and contrast using Fiji.

3. Results and discussion

3.1. Tissue formation and shape analysis

hPDC cells were seeded in low-adherent well plates in various concentrations and left to self-assemble. The number of seeded cells ranged from 5k to 100k, which resulted in the formation of well-defined spheroids for the low regime, transitioning in shape to an ellipsoid-like tissue, (i.e. meaning that the ratio of the diameters D_L/D_T decreases as seen in Fig. 1a) as the size increased (Fig. 1a). Here we define the sphericity of microtissues as the ratio of the two vertical diameters D_T (transverse to the loading direction) and D_L (longitudinal to the loading direction) (Fig. 1a). We note that all the tissues regardless of their D_T and D_L, are circular when viewed from the top view.

Surface tension plays a pivotal role in defining the shapes of biological tissues [40]. We need to distinguish between the cell-liquid surface tension which is developed at the interface of the tissue and the surrounding medium, and the surface tension of the tissue which stems from the internal cell-cell and cell-ECM interactions, forces generated by the contractility of the cell cortex and the elasticity of the cell cortex [41]. Herein, we focus on the role of the interfacial surface tension (IST) on the mechanical properties of the tissues which is extensively studied [19] and there is analytical solution for its estimation [11] which is adopted herein. The shape of a tissue is influenced by the balance between intracellular and extracellular forces acting on the cells. These forces are regulated by surface tension, which works to reduce the exposed surface area of cell clusters while enhancing their cohesive strength [42]. Surface tension becomes significant when it falls below the capillary length (l_s), a scaling factor relating surface tension and gravity, given by $l_s = \sqrt{(\gamma/\rho g)}$ where γ is the surface tension, ρ is the mass density of the tissue and g is the gravitational acceleration [43]. The capillary length can also be related to the modulus of a solid with surface tension by the relation $l_s = \sigma/E$, where σ is the surface tension and E is the Young's modulus. Since the surface tension varies with tissue size as discussed in detail below, theoretical estimations using the existing theories are case sensitive and refer to specific conditions. Instead, we provide values for l_s based on the sphericity transitions with tissue dimensions and we discuss further this point below. In our case, it seems that the capillary length is estimated to be ~400 μm for the cell aggregates and ~1 mm for the cartilaginous microtissues based on the sphericity transitions in Fig. 1e. In Fig. 1e the sphericity ratio as measured by optical inspection by a side view camera is given for the range of the examined dimensions for cell aggregates and cartilaginous microtissues, which shows that as the tissue size increases, the sphericity gradually decreases. In Fig. 1f, the IST measured at rest (i.e. at 0 % of compression and details for the calculations are given below) depends on the sphericity ratio and increases as the tissue shape transits from ellipsoidal to spherical.

The resulting shapes presented in Fig. 1 suggest that gravity does not affect the formation of nearly perfect spheroids with diameter less than 400 μm for the cell aggregates. With the increase in the number of seeded cells, gravitational forces result in outbalancing the internal cohesive strength between cells and surface stresses, and the sphericity gradually decreases. This is similar to the flattening of liquid droplets due to gravity for increasing droplet size which has been clearly observed in other studies [44,45]. Elsewhere [46], tissue surface tension in relation to the sphericity of the tissue was measured experimentally. By incorporating magnetic nanoparticles into the tissues, the initial spheroid tissues deformed by the application of a magnetic field which transforms the tissue shape from spherical to ellipsoid as in Fig. 1e. The results showed [46] that the perfect spheroids possess higher surface tension than the deformed tissues, and the surface tension decreased with the decrease in the sphericity. This is also observed herein, and as the results in Fig. 1f show the surface tension decreases with the decrease in sphericity of the samples, and with the increase in D_T as seen in Fig. 1f. In our case, the only additional forces when tissue size increases stem from gravity, thus the analysis strongly suggests that the shape evolution is determined by the balance of surface tension effects and gravitational forces. Cell contractility also plays a crucial role in the shape formation of a tissue by generating cellular forces which contribute to the development of surface stresses [47,48]. In line with this, inhibition of myosin contractility for the cell aggregates by treatment with the Rho-associated protein kinase (ROCK) inhibitor Y-27632[49], resulted in decreased sphericity and IST as presented in Fig. S8.

Cell aggregates of varying dimensions differentiated in chondrogenic medium over a period of up to 21 days yielded cartilaginous microtissues. By day 21 these microtissues exhibited phenotypic properties resembling those of (pre)hypertrophic chondrocytes, as observed during the cartilage to bone transition during long bone development and defect regeneration, a process known as endochondral ossification [2]. The sphericity of large cartilaginous constructs markedly exceeded that of undifferentiated aggregates (Fig. 1d and e), which is also reflected in increased IST values for these samples as compared to the cell aggregates (Fig. 1f). We attribute the increased sphericity to the presence of collagen that is secreted during differentiation. The collagen fibers are stiff and contribute to the increased rigidity of the cartilaginous microtissues, and possibly stronger internal cohesion to the cell-ECM bonds [49].

3.2. Non-linear mechanical properties of cell aggregates

Previous works estimated the Young’s modulus of cellular spheroids under compression using either the Hertz model [9] or by treating the spheroid as a viscoelastic solid with the strain defined as the relative change in the radius in the direction parallel to the applied load and the stress in this direction [6,7]. Here, we adopted the latter approach since we were most interested in the mechanical response under large deformations for which the Hertz model is not applicable. The contact area between the plate and the tissue can be directly measured by optical inspection with sub-micron resolution for the whole loading history, thus the calculation of stress-strain curves from the measured force is straightforward (Fig. 2). Details on the calculations for the Young's modulus are given in the experimental section and in SI.

The Young's modulus (E), defined as the slope of the stress-strain curve at 0 % strain, was strongly dependent on tissue dimensions, and was empirically described by a power-law relationship $E \propto D_T^{-2 \pm 0.2}$ ($R^2 = 0.75$) (Fig. 3a). We note here that the Young's modulus obtained is a measure of resistance provided by the cytoskeletal structure and surface tension effects and can thus be considered an effective Young's modulus. We attribute this scaling relation of E to surface tension effects, which can be estimated based on the radii of the tissue in the deformed state and the applied force [39]. Surface tension effects are expected to be eliminated for tissue dimensions over the capillary length [40]. In agreement with this, E (and IST as presented below) stabilized over D_T ~800 μm (Fig. 5a), close to the upper limit of the capillary length as explained above.

Surface tension has been found to be pressure [15] and size [50] dependent, thus calculations performed at 10 %, 30 % and 50 % of compressive strain (or 10 %, 20 % and 30 % for samples that tested until maximum compression of 30 %) and the calculations cover the range of dimensions described above. In this regime the stress-strain response is linear, and the strain stiffening is avoided. In Fig. 3 the IST at rest in function of the D_T is presented for all tissue types. The results show that the IST varies with the level of compression and diameter, confirming the presence of a surface modulus with similar effects found in other studies for soft solids [51]. The IST increases with the increase in compressive strain and the IST at 0 % compression is estimated by extrapolation [15]. The same scaling of IST with the D_T is observed as for the case of the Young's modulus. The variation of the Young's modulus and the IST with the size is statistically significant as seen by the p-values ($p < 0.001$ for all cases), as well as between time points. In order to explain the observed size effect, we resort to the scaling of the forces that act on the tissues in relation to their dimensions. The forces due to surface tension scale with the length of the tissue as $F_{ST} \propto L^{-1}$, rather than the area as reflected by the unit of surface tension which is N/m, while the bulk forces scale with the volume of the tissue as $F_{\mathrm{Bulk}} \propto L^{-3}$. Thus, it follows that the balance of forces scale with L^{-2} across scales. Some deviations are expected because the tissue shape transits gradually from spherical to ellipsoidal and the volume depends on the vertical lengths as the dimensions of the tissue increase. For the cartilaginous microtissues the scaling of the Young's modulus shows small quantitative differences, although it also follows an exponential dependence. For these tissues, the fitting is affected by the larger average modulus for larger D_T (>500 μm) and it is discussed in the next section.

The same scaling of IST and Young's modulus with D_T implies a linear relation between the two quantities, which is confirmed by plots presented in Fig. 3. The correlation can be approximated by linear fitting in the form of $y = Ax + B$. As explained above, the measured Young's modulus is the result of the contribution of the elasticity of the bulk tissue and the effect of surface tension. Thus, it can be described by the following equation:

$$E = E_C + \frac{\sigma}{l} \quad (1)$$

where E_C is the Young's modulus of the core (or bulk) of the tissue, σ is the IST, while l corresponds to the slope and is an inverse length measure. The size effect is incorporated in the surface tension term (σ) as a function of the tissue diameter (or radius). All quantities of equation (1) can be estimated by fitting the experimental results of σ-E (σ here is the IST) presented in Fig. 3 for each tissue type, and the values obtained for E_c and l are given in Table 1. We note that in other studies, the length of the second term of the right side of equation (1) is the capillary length for the case of soft solids [52], the radius of the tissue [53] or an equivalent thickness for the radius of a single cell with estimated value of 10 μm in Ref. [24]. Since here we account for the interfacial surface tension between cells and the surrounding medium, it is plausible that this length represents a thickness at the tissue's periphery which is affected by the forces acting at the interface cells-liquid. This simple empirically derived model offers insights on the observed mechanical response. The elasticity of the bulk tissue increases with the presence of collagen which is plausible to the significantly higher stiffness of the collagen, and is reflected in the increase observed in the E_C. The effect of IST is captured by the surface modulus given by the second term of equation of 1, and the size dependence of the modulus stems from the scaling of the IST.

An effective surface tension can in some cases reflect the formation of a "tensile skin" or "ring" on the periphery of the cell aggregates that is stiffer than the core of the spheroid [23]. Previous results estimated the stiffness of the ring to be 2 to 30 times higher than the stiffness of the core [23,24]. In our case, this modulus can be estimated by σ/l, and represents a surface modulus [51]. The values of l obtained from the fittings shown in Fig. 3, are in the range 8–15 μm (Table 1), in very good agreement with reference [24]. We notice that l decreases with differentiation time towards cartilaginous phenotype. This is clearly seen in the high-resolution images for the tensile ring in Fig. 4 (third row). We found that the cells on the periphery appeared to be stretched and formed a ring with thickness $t_{\text{Ring}} \sim 20 - 40\ \mu\text{m}$ (considering the actin fibers stretched parallel to the periphery of the tissue) for the cell aggregate, and decreases with differentiation time. Good agreement between theory and experiments for the cartilaginous microtissues, and small deviations for the case of cell aggregates (Table 1 and Fig. 4). Thus, the analysis strongly suggests that l is the thickness of the tensile ring. With increasing aggregate size, the interior volume scales as $V \sim (R - t_{\text{Ring}})$ [3]. Since the t_{Ring} is constant (as seen by comparing the images of Fig. 4 and Fig. S1 for tissues with diameters on the low and high regimes) its effect on the effective modulus becomes negligible as the size of the aggregates increases, and the core stiffness can be considered this of the large aggregates with average value ~1 kPa. This value derived from cell aggregates with D_T > 800 μm where the modulus stabilizes and forms a plateau. The bulk elasticity (E_b) of a tissue has been found to be dependent on surface tension linearly, i.e. $E_b \propto R^{-1}$, while the surface elasticity (E_s) is correlated to the radius of the spheroid by $E_b \propto R^{-2}$ [24]. This analysis agrees with the scaling behavior observed in the present experiments (Figs. 3 and 5) and suggests that surface tension has a pivotal role in determining the effective stiffness scaling observed in this work.

We were surprised by the exceptional resilience of hPDC cell aggregates, which sustained compressions up to 95 % without failure. To quantify stiffness in the non-linear regime, we estimated the tangential stiffness (slope of the stress-strain curve) at the maximum compressive strain (Fig. 3b). This tangent stiffness reached values up to one order of magnitude higher than the initial linear modulus. The corresponding maximum stresses provide a lower bound for tissue strength (Fig. 3c). A size effect was also evident for the tangent stiffness, with values up to three times higher for the smaller tissues compared to the larger tissues at equivalent peak compressions.

Strain stiffening in epithelial cell monolayers results from the straightening of intermediate filaments (IFs), known for their intrinsically strain-stiffening behavior [16]. In our case, we anticipate vimentin IFs, which are highly expressed in hPDCs, may similarly provide strain stiffening [54] under compression, and their organization within the cell aggregate structure is seen in Fig. 6. However, additional work would be required to confirm the role of the IF cytoskeleton in the strain stiffening observed in this study.

3.3. Non-linear mechanical properties of cartilaginous microtissues

Cartilaginous microtissues were stiffer than their cell aggregate counterparts and also exhibited a pronounced dependence on size (Fig. 5c and d, g, j). The scaling of Young's modulus with D_T was qualitatively similar to that observed in cell aggregates, with exponent values are −0.61 for day 7, −1.01 for day 14, and −1.27 for day 21, as shown in Fig. S4, the R^2 values drop to ~0.5. The lower values of the exponent show a less abrupt increase of the Young's modulus with decreasing dimensions. For D_T > 400 μm the Young's modulus is 3–4 times higher than the corresponding values for the cell aggregates, as obtained from equation (1). Below this D_T the average increase in the Young's modulus is lower that the increase in the higher D_T regime, resulting in lower values of the exponent m.

We note that after 7 days in chondrogenic medium the microtissue sizes modestly decreased as the differentiation time progresses, in agreement with previous results where apoptosis was observed [2]. A major structural difference between cell aggregates and cartilaginous microtissues is the secretion of collagen fibers of type I and II (Fig. 4) [2]. The cartilaginous phenotype is confirmed by histological staining for collagen type I and II, as well as Safranin-O and alcian for tissues of varying size as presented in Figs. S1 and S2, which confirms the cartilaginous phenotype of the differentiated tissues. In some cases, small structural imperfections like the lack of periphery ring and deviations from perfect spherical shapes are observed as for example for the cartilaginous microtissues day 14 and 21 seen in Fig. 4. Such imperfections can also contribute to the variation of the mechanical properties. As the differentiation time progresses, the collagen fibers mature and the whole tissue becomes more compact. This is seen in Fig. S4 with optical images taken for the same tissues during differentiation for the examined time-points and for various dimensions of samples, which show that this phenomenon occurs independently of the size of the samples.

The increase in Young's modulus for the cartilaginous microtissues is most pronounced for larger D_T, where the average E increases to ~5 kPa for D_T > 800 μm (for day 7 and 14), five times higher than that of cell aggregates of similar D_T. In this regime, the sphericity of the cartilaginous microtissues was twice that of the cell aggregates' (Fig. 1e), suggesting that the stiff collagen network makes the tissue more able to resist gravitational forces. It is also plausible that a shift from primarily cell-cell junctions to cell-ECM adhesions contribute to the increased tissue integrity [49]. This is also shown by the values of the IST presented in Fig. 3, that follow the same trend as the Young's modulus. The increase in IST along with the resistance to compression provided by the stiff collagen network results in overall mechanically robust and much stiffer tissues. The peripheral ring comprised of actin cables was also present in the cartilaginous microtissues (Fig. 4a). This ring also contributes to the abrupt increase of stiffness for D_T < 400 μm, as it is suggested by the theoretical model.

Beyond 60 % compression, all samples showed a sharp non-linear increase in stiffness (Fig. 2), and increased tangent moduli as compared to cell aggregates (Fig. 3). It is possible that collagen fibers provide strain stiffening similar to the role of IFs in the cell aggregates, as collagen shows a strain stiffening response under large deformations [55,56]. Vimentin IFs are also present in the structure of the cartilaginous microtissues, but their structure is subjected to changes and the images show that remodeling occurs with the secretion of collagen (Fig. 4). Further studies are needed in order to identify their distinct roles under compressive load. For mature cartilaginous microtissues (day 21), the tangent stiffness reached ~200 kPa, clearly demonstrating stiffening by the presence of the collagen and reflecting the exceptional mechanical properties and resilience of our *in vitro* tissues.

3.4. Implications of size dependent mechanics for tissue engineering

Our *in vitro* tissues demonstrated remarkable resilience under compression. This resilience makes them potentially well-suited for tissue engineering [2]. To our knowledge, the compressive moduli measured herein are among the highest reported for *in vitro* model tissues [7–12]. The size-dependent mechanical properties have potentially important implications for tissue engineering. Previously, mature cartilaginous microtissues have been implanted in mice and successfully healed long bone fractures [2]. In these experiments, a large number of spheroids <250 μm in diameter were implanted, and subsequently fused together. Our results demonstrate that the effective stiffness of cartilaginous microtissues scales inversely with size. Hence, microtissue size control can become a target parameter modulating mechanical properties of implanted tissues.

The effect of surface tension on cell and tissue mechanics is the subject of several studies [24]. However, it remains to be determined whether an effective surface tension is a common feature of living tissues, and whether its effects follow universal scaling laws. For example, decreasing stiffness with increasing dimensions might not necessarily be universal, as the stiffness of at least some cancer cells and tumors was reported to decrease with a decrease in dimensions [57]. The general question of how the mechanical properties of tissues scale with size thus presents an important opportunity for future research.

3.5. Stress relaxation and viscosity

To examine stress relaxation dynamics, tissues were compressed to 30 % and 50 % without altering the loading rate, and the resulting forces were recorded over a period of 1 min (Fig. 5). For cell aggregates, the stress dropped abruptly and within 5 s 50–70 % of stress is dissipated. In contrast, for cartilaginous microtissues only 20–25 % of peak stress dissipated in the same timeframe. Although stress dissipation accelerated with increased initial compression in all cases, the overall amount of dissipated stress remained relatively constant on average.

The initial stress dissipation can be fitted with a power-law function, which follows $At^{-\alpha}$, where A is amplitude, t is time, and α is an exponent that sets the timescale for stress dissipation [58] (Fig. 8a). A similar exponential response has been noted for other tissues such that of stress decreases monotonically and stabilizes over time [24]. In contrast, after an initial relaxation, stress in our tissues showed highly oscillatory behavior (Fig. 7), as will discussed below. In all cases, higher peak compressions led to an increased α, which reflects faster stress dissipation. Although size effects have been noted in the viscoelastic response of cancer cells and tumors [32], no significant differences observed in the relaxation times for cell aggregates and microtissues of varying size. The value of α decreases linearly with differentiation time, likely reflecting a decrease in stress dissipation due to the presence of collagen (Fig. 8f).

In order to gain further insight into the viscoelastic properties of the tissues, we estimated the viscosity (η) for all cases using the Maxwell approximation and the results are presented in Fig. 9 (details about the calculations are given in the methods). As seen in Fig. 9, the viscosity is also size dependent due to its direct relation to the Young's modulus and is found to be statistically significant ($p < 0.001$ for all tissue types). The scaling of the viscosity with D_T is qualitatively similar to the scaling of the Young's modulus and surface tension. A dramatic increase is observed in viscosity with differentiation time, which is on average 32.69 kPa s for the cell aggregates and for the cartilaginous microtissues day-21 the average value increases more than five-fold to 172.95 kPa s. The viscosity does not depend on the strain magnitude of the compression, showing that our tissues can be described by linear viscoelastic theory which validates the approximation of the Maxwell approach for viscoelastic materials. This increase reflects the transition of the tissue from a fluidic behavior of the cell aggregate to a more solid material response, which is physiologically relevant since this is the intermediate form of the tissue that eventually forms bone.

Following the initial relaxation under compression, mechanical stresses showed an oscillatory behavior for both cell aggregates and cartilaginous microtissues (Fig. 7). These oscillations were suppressed under higher compressions. This oscillatory response was confirmed by optical inspection of the tissues during the stress relaxation test (movie S1). The stress relaxation response of an elastomer was also tested as a negative control and showed a purely elastic response within the same time frame (Fig. S7). The oscillatory response deviates from the behavior of viscoelastic solid or fluid, showing that there is another dynamic mechanism that tissues respond to external mechanical stimuli. Conceptually similar periodic oscillations have been observed in multiple developmental processes and potentially provide a means by which developing tissues can probe their physical environments [59,60]. More specifically, one of the most examined oscillations mechanisms is gastrulation in drosophila which is governed by the actomyosin network with oscillations occurring with periodicity on minutes time scale [59, 60]. It is possible that a similar mechanism of collective mechanosensing may also be operative during the healing of bone fractures [2].

The oscillatory response after the first abrupt relaxation of stress points to active force generation by actin and myosin, which has been observed in other studies with oscillations occurring on the minute time scale [61–64]. The mechanisms underlying such oscillations are complex and context dependent. The origins of force oscillations that we observe, while potentially interesting, are best left for future investigations. For the cell aggregates we observe periodicities of ~6 and ~10 s for the force oscillation. These values coincide with the turnover times of myosin, actin and cytoskeletal crosslinkers [65]. We therefore quantified the stress relaxation response of cell aggregates treated with the Rho-associated protein kinase (ROCK) inhibitor Y-27632, which disrupts the myosin contractility (Fig. 10a) [66]. This treatment resulted in a significant increase in the relative magnitude of the force oscillations, as well as a shift in periodicity to >10 s. A shift in the periodicity was also observed for cartilaginous microtissues, noting that the chondrogenic medium contains Rho-inhibitor. As seen in Fig. 10b and c, inhibition of the myosin contractility does not affect the initial stress dissipation as the values of the fitting exponent α overlap with the cell aggregates' values, while the viscosity decreases significantly on average values compared to the cell aggregates because of the decrease in the Young's modulus. Thus, treated tissues show more fluidic behavior compared to the cell aggregates and cartilaginous microtissues.

3.6. Toward mechanically informed design of modular tissue engineered implants

Modular skeletal tissue engineering employing the use of cartilaginous microtissue building blocks is a rapidly growing domain [67,68] due to its promising results and biomimicry of the cartilage to bone transition observed during fracture healing [69]. Cartilaginous microtissues derived from various adult progenitor cell types such as bone marrow MSCs [4,70,71], adipose derived MSCs [36,72] and hPDCs [72, 73] have been to date validated for their capacity to undergo endochondral ossification upon implantation in small animal models. Some of these studies demonstrated even the potential of this strategy for regeneration of critical size tibial [35] and cranial defects [35]. Moreover, cartilaginous microtissues can be merged with biomaterials such as melt electrowritten meshes [74] and hydrogels [75] in order to produce microtissue-hydrogel hybrids. In both scaffold free and hybrid implants the mechanical properties of the microtissues will define the overall mechanical properties. The rapid stress relaxation observed in this study has important implications for cartilage tissue engineering. The microtissues formed by hPDCs recapitulate the phenotypic transitions that occur during endochondral ossification and can robustly form bone upon implantation [2]. In principle, a scaffold-free approach allows a more natural differentiation process that is not reliant on the implantation of an engineered scaffold. However, one design consideration for this approach is the selection of the appropriate dimensions of the microtissue building blocks. Our results indicate that increasing the size of cartilaginous microtissues reduces their stiffness. The bottom-up fusion of smaller microtissues to create larger granular tissue constructs *in situ* provides a possible means of addressing this limitation [76]. However, further studies are needed to determine how the mechanical properties of the emerging bioassembled tissue evolve during this fusion process. Conversely, for approaches that involve the use of scaffolding, it is desirable to match the materials properties of the scaffold as well with its degradation with those of the native tissue. Hydrogels that exhibit fast stress relaxation have been shown to promote chondrogenic differentiation of adult progenitors as compared to hydrogels with elastic response [77–79], and to promote bone formation *in vivo* [80]. These findings are consistent with the rapid, though limited stress relaxation exhibited by the cartilaginous microtissues studied here.

The mechanical cross talk between cartilaginous-implants and load at the defect site has been seen to play a pivotal role in fracture repair [81], hence this study provides substantial input in selecting microtissues with appropriate mechanical properties. For clinical translation scalable technologies able to produce high number of mechanically robust cartilaginous microtissues are currently being explored [82]. Moreover, bioprinting technologies tailored for handling microtissues either through extrusion [83], laser [84] or aspiration mechanisms [85] are currently designed in order to provide high-throughput and precision for microtissue-based implants. Taken together these biomanufacturing technologies could further obtained mechanical parameters as design input for the bioprinted of mechanically tailored implants according to defect location, size taking into account even personalized aspects.

4. Conclusions

In this study, we observed a clear dependency of mechanical properties on tissue size, characterized by a scaling law $E \propto D^m$. In addition, cartilaginous microtissues were extremely resilient, and could sustain over 90 % of compression without mechanical failure. These tissues exhibited strain stiffening under large compressions, with a multi-fold increase in the tangent stiffness. Finally, microtissues exhibited a viscoelastic response on short timescales of a few seconds, and actomyosin-driven oscillations in residual stress on the minutes timescale. The microtissues examined here thus recapitulate properties of the actual *in vivo* conditions such as strain stiffening under large deformations and an initial, fast stress relaxation, similar to the response observed in explants of human articular cartilage [86]. The results also reveal a dynamic response to external mechanical compression that is reminiscent of mechanical oscillations observed in multiple developmental contexts [42]. The findings suggest these microtissues are well-suited as models for bone tissue engineering as well as organ-on-a-chip applications for studying cartilage mechanics. Finally, our work underscores the importance of surface tensions, and more broadly tissue size, in determining the mechanical properties of living tissues both in the context of embryonic development and in tissue engineering.

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

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Data availability

Data will be made available on request.

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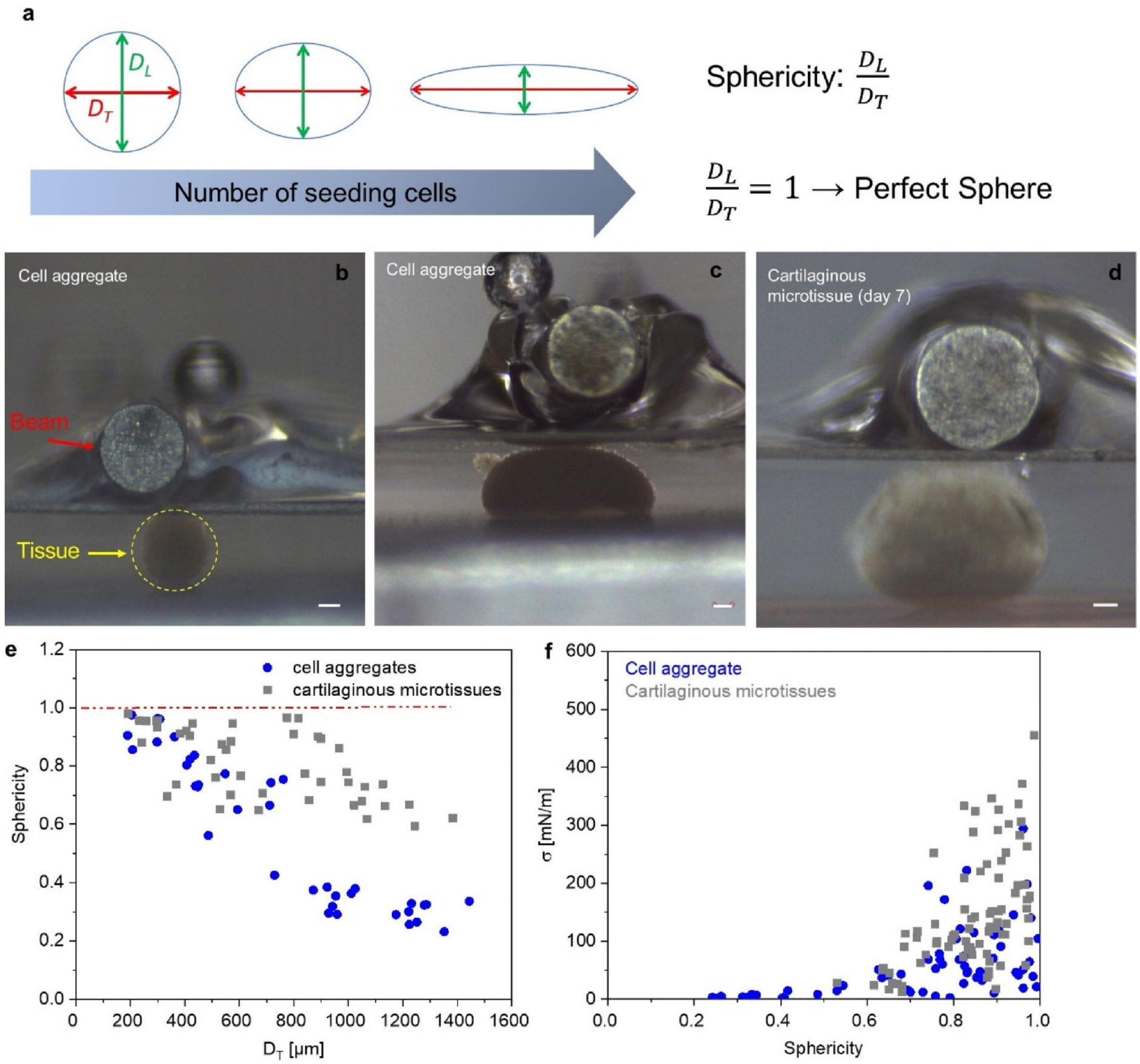

Fig. 1.

Tissue shape formation across scales and sphericity. (a) Schematic of the resulting tissue shapes. Definition of transverse (D_T), longitudinal diameters (D_L) and sphericity. (b, d) Side view images of small (b) and large (c) cell aggregates, and (d) a large cartilaginous microtissue prior to compression. The yellow dashed line in b captures the tissue, while the red arrow indicates the beam that applies compression. (e) Sphericity ratios for cell aggregates and cartilaginous microtissues as measured from side-view images across the examined range of dimensions (cell aggregate $n = 37$, cartilaginous microtissues $n = 44$). The red dashed line is a guide to the eye corresponding to perfect sphericity (i.e $D_T = D_L$). (f) The tissue surface tension at rest in function of the sphericity for cell aggregates ($n = 60$) and cartilaginous microtissues ($n = 83$). (f) Values of the interfacial surface tension in function of the sphericity of cell aggregates and cartilaginous microtissues. Scale bar = 100 μm (b–d).

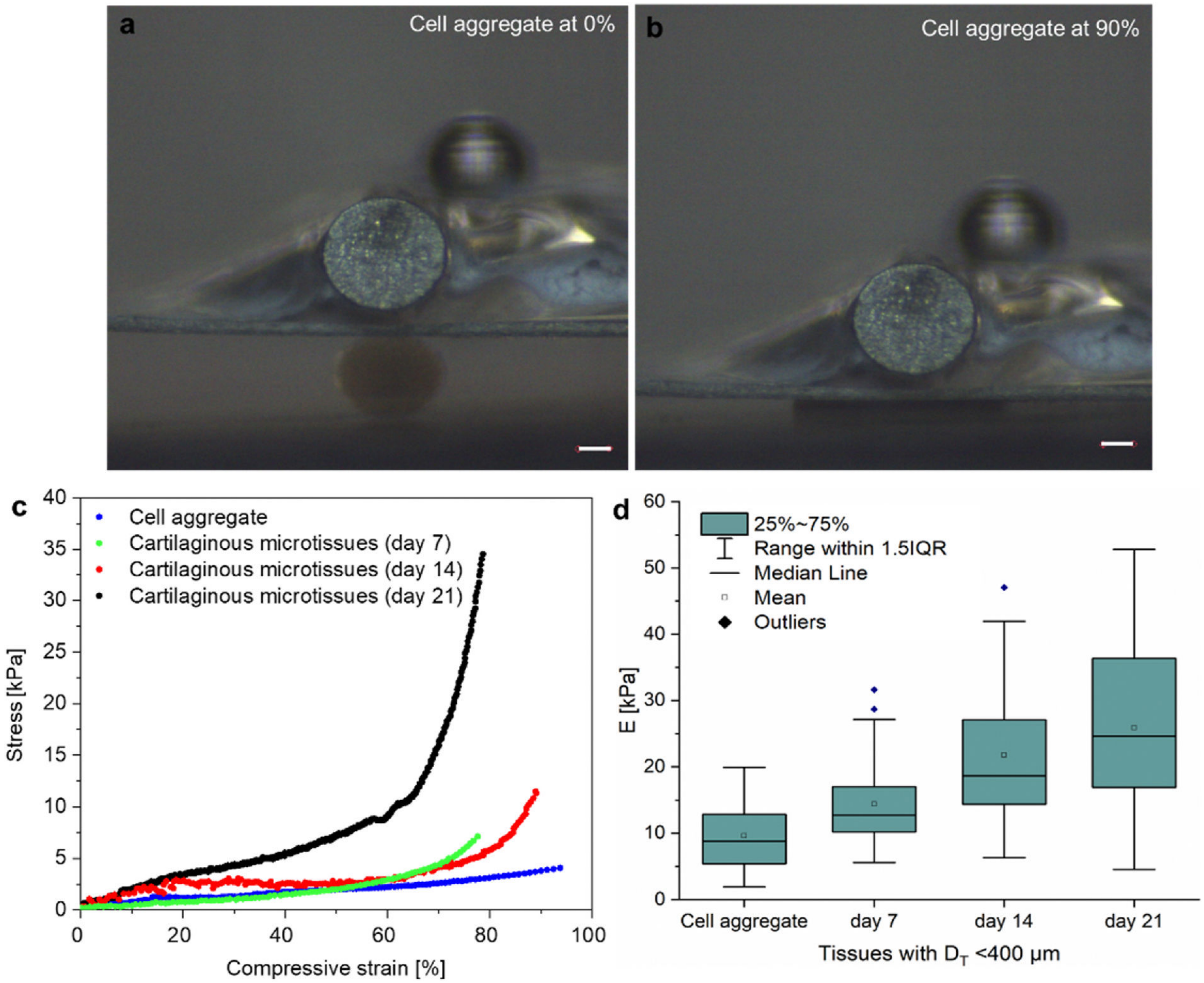

Fig. 2.

Cell aggregates and cartilaginous microtissues are resilient under large compressive strains. (a, b) Snapshots from a compression experiment for a cell aggregate at strains 0 % and 90 %, respectively. (c) Stress-strain curves for cell aggregate and cartilaginous microtissues for various time points of differentiation. (d) Average Young's modulus of tissues with D_T < 400 μm (cell aggregates: *n = 46*, cartilaginous microtissues: $n_{day\text{-}7} = 34$, $n_{day\text{-}14} = 42$, $n_{day\text{-}21} = 52$).

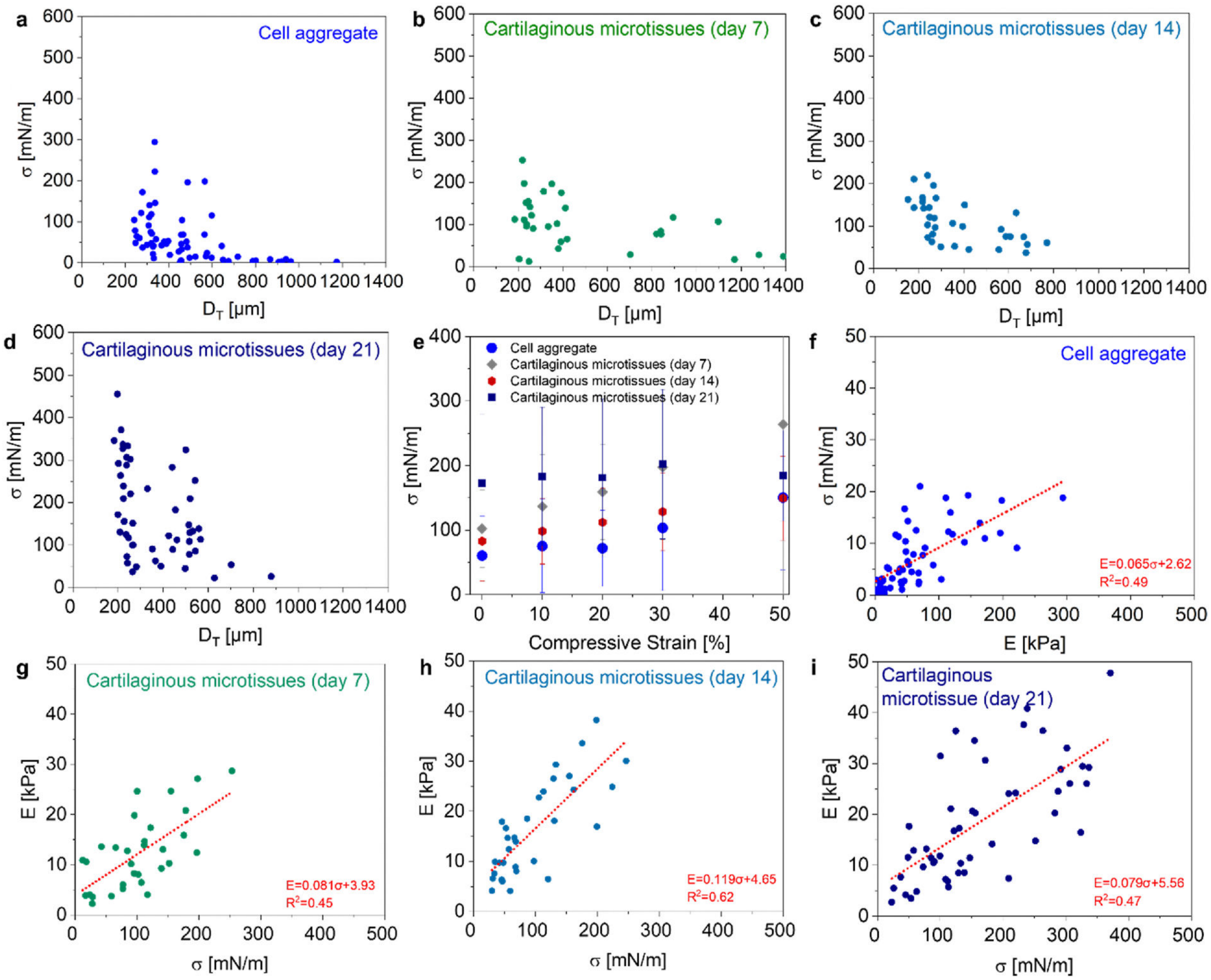

Fig. 3.

Interfacial surface tension is size dependent and is linearly related to the Young's modulus. The tissue surface tension at rest in function of the D_T for (a) cell aggregate (n = 59, $p < 0.001$), (b) cartilaginous microtissues day 7 (n = 30, $p < 0.001$), (c) cartilaginous microtissues day 14 (n = 31, $p < 0.001$) and (d) cartilaginous microtissues day 21 (n = 50, $p < 0.001$). (e) The average values of interfacial surface tension at various magnitudes of compression. The interfacial surface tension at rest in function of the Young's modulus for (f) cell aggregate (n = 59), (g) cartilaginous microtissues day 7 (n = 30), (h) cartilaginous microtissues day 14 (n = 31) and (i) cartilaginous microtissues day 21 (n = 50). The dashed red lines in (f–i) represent linear regression and the corresponding slopes are in microns.

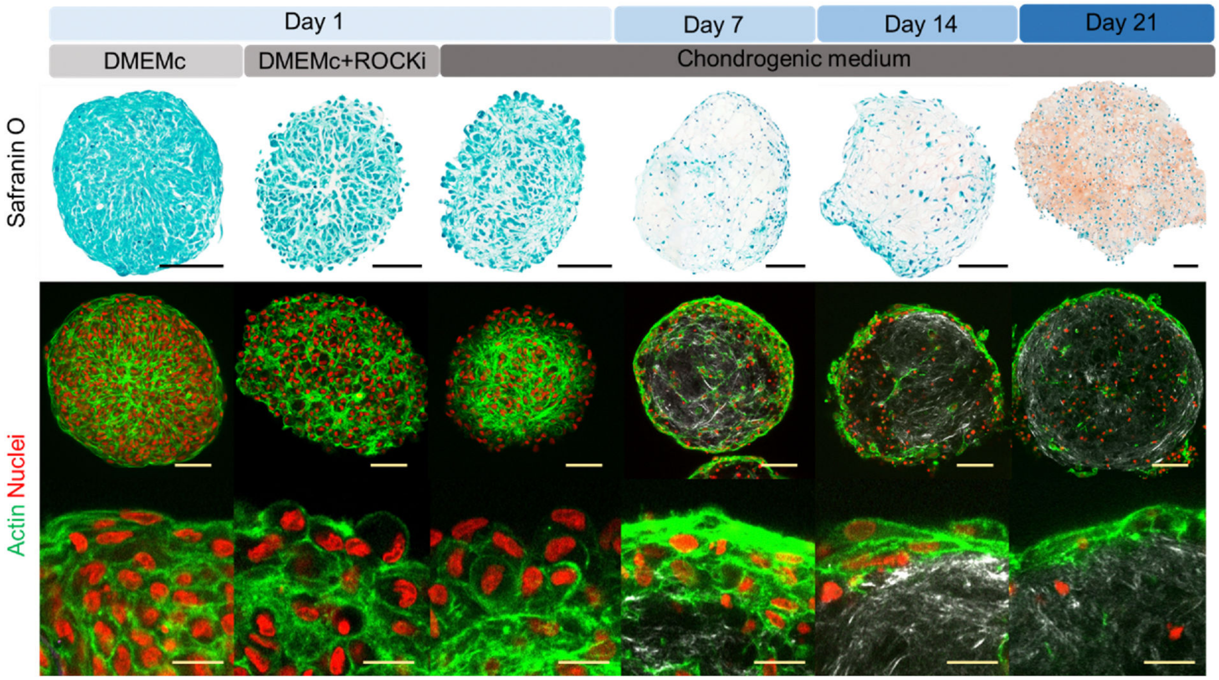

Fig. 4.

Tensile ring formed by actin cables at the periphery of the tissues. First row: Histological analysis of cell aggregate, cell aggregate treated with Y27632 and cartilaginous microtissues at various time points of differentiation stained with Safranin O. Second row: Immunofluorescent images stained for actin (green), nucleus (red) and collagen visualized with second harmonics (white) for all examined tissue types. Third row: high magnification images capturing the periphery of the tissues and the “tensile” ring with stretched actin cables for all tissue types. In all cases the seeded number of cells is 5k. Scale bar: 100 μm for the first and second row and 20 μm for the third row.

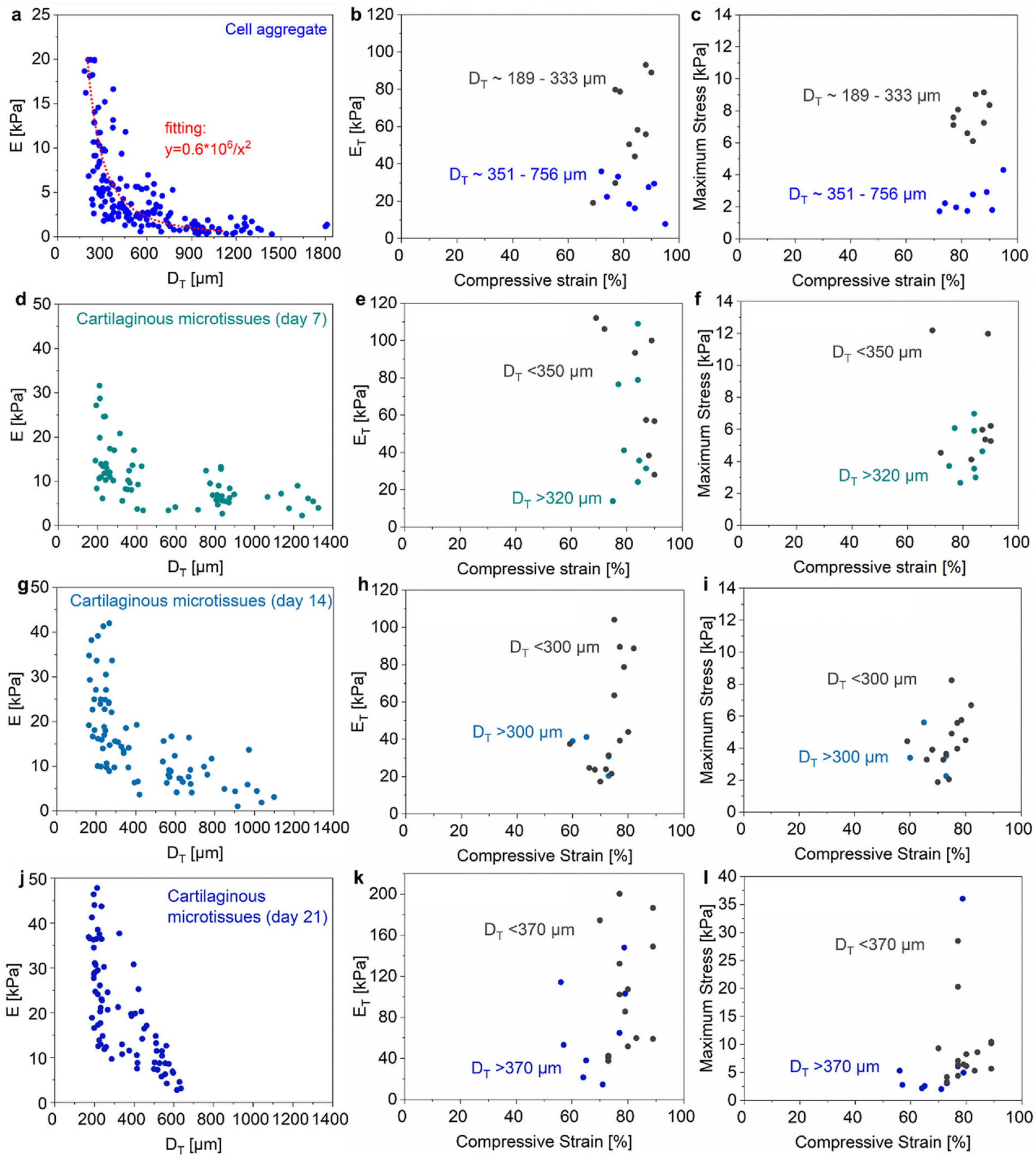

Fig. 5.

Cell aggregate and cartilaginous microtissue size affects effective modulus and resiliency under large strains. (a–c) Mechanical properties of hPDC cell aggregates: (a) Effective Young's modulus versus the diameter transverse to the applied compression (*n = 167, p < 0.001*). (b) The tangent stiffness at the peak compressive strain (*n = 18*). (c) The maximum stress at the peak compressive strain (*n = 18, p*). (d–f) Mechanical properties of cartilaginous microtissues on day 7 of differentiation: (d) Young's modulus (*n = 67, p < 0.001*), (e) tangent stiffness (*n = 16*), (f) Maximum stress (*n = 16*). (g–i) Mechanical properties of cartilaginous microtissues on day 14 of differentiation: (g) Young's modulus ($n = 78, p < 0.001$), (h) tangent stiffness ($n = 20$), (i) Maximum stress ($n = 20$). (j–l) Mechanical properties of cartilaginous microtissues on day 21 of differentiation. (j) Young's modulus ($n = 79, p < 0.001$), (k) tangent stiffness ($n = 22$), (l) Maximum stress ($n = 25$).

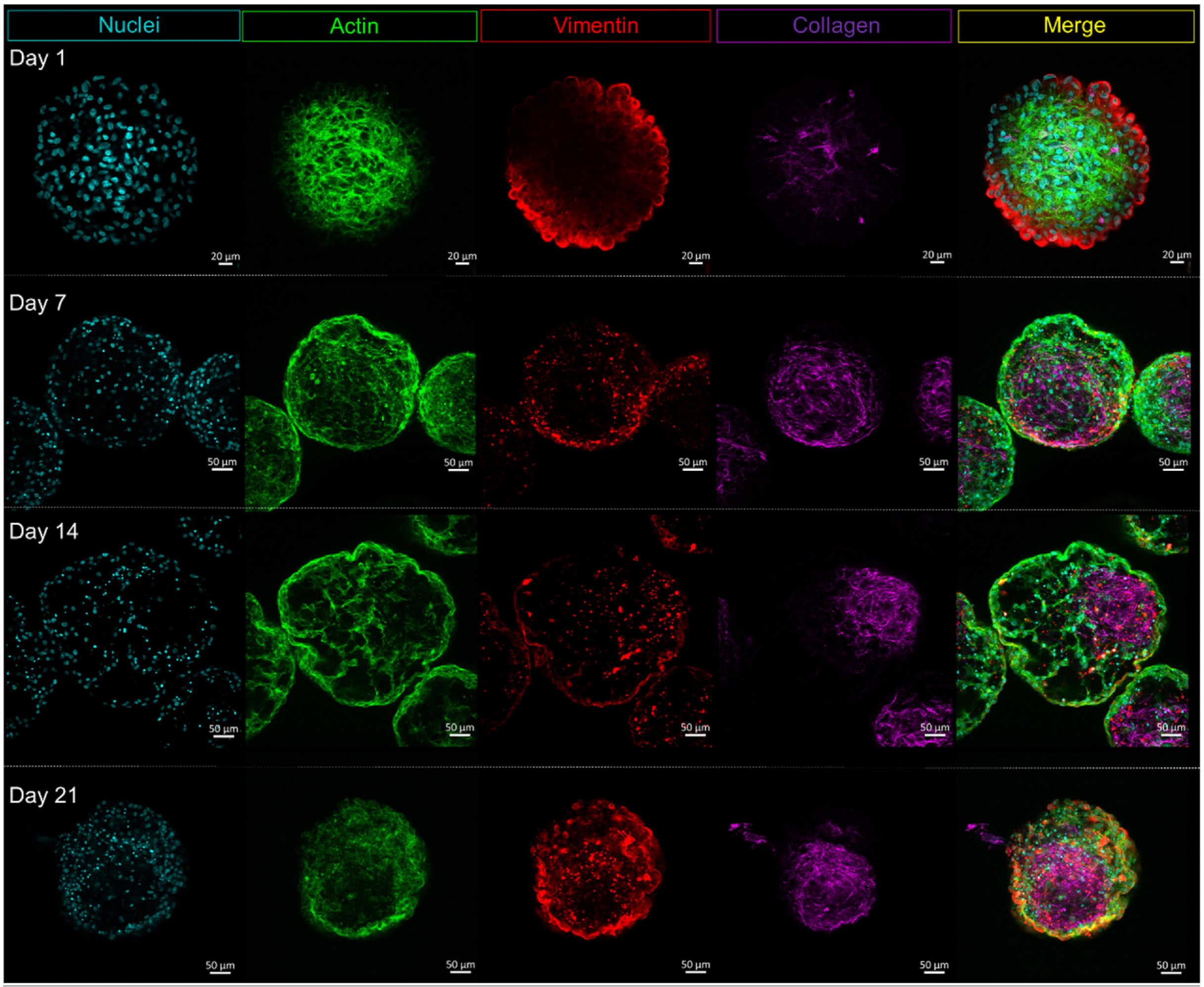

Fig. 6.
Matrix remodeling with secretion of collagen during differentiation towards cartilage. Immunostaining imaging of tissues stained for nucleus, actin and vimentin for various time points of differentiation. Collagen visualized by second harmonics.

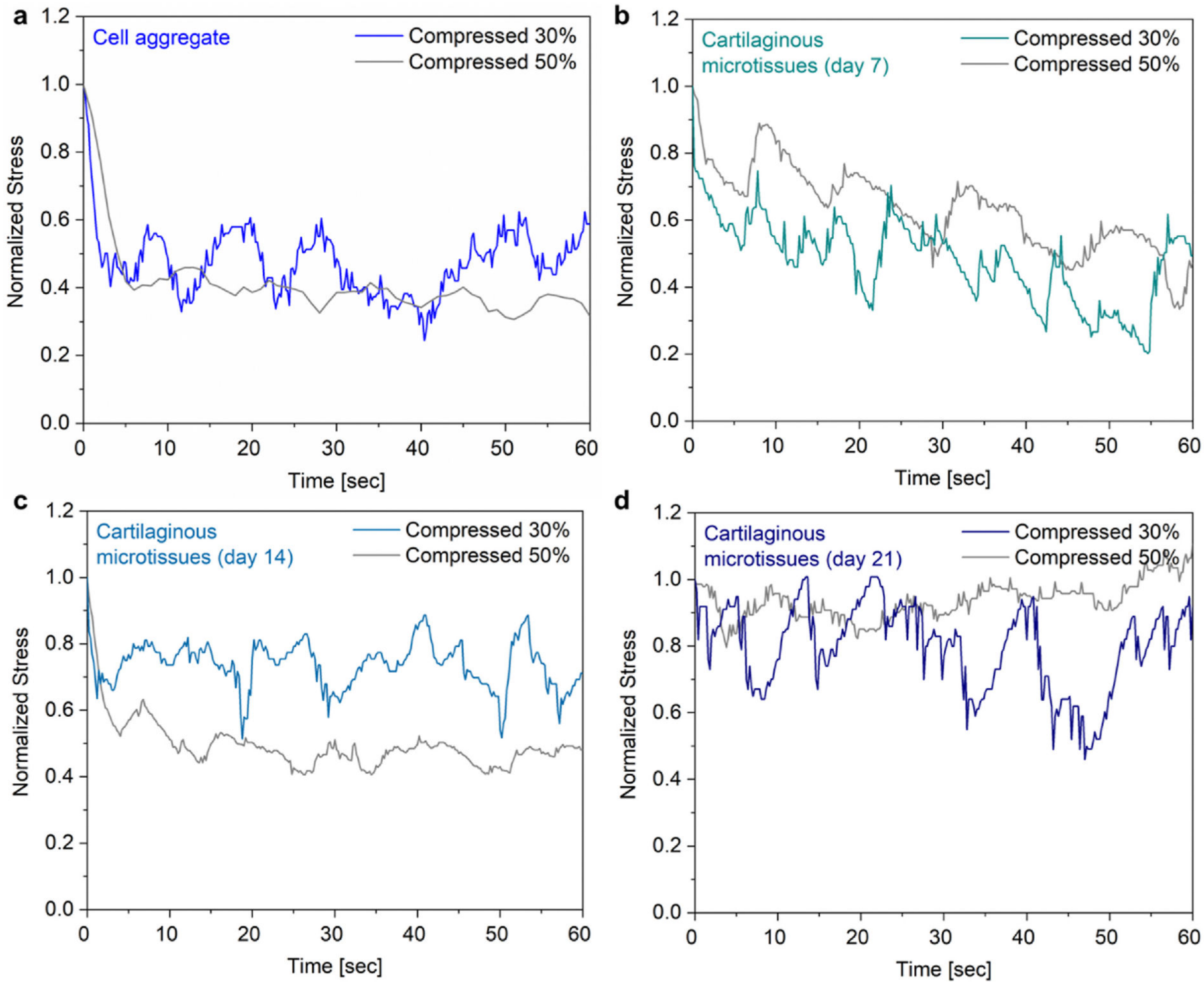

Fig. 7.

Tissues dissipate stress rapidly on a short time scale followed by periodic oscillations. Stress relaxation response (stress vs. time) for peak compressions of 30 % and 50 %, corresponding to (a) cell aggregates, (b) cartilaginous microtissues day 7, (c) cartilaginous microtissues day 14 and (d) cartilaginous microtissues day 21. Stress values are normalized to the peak applied stress for comparison between cases.

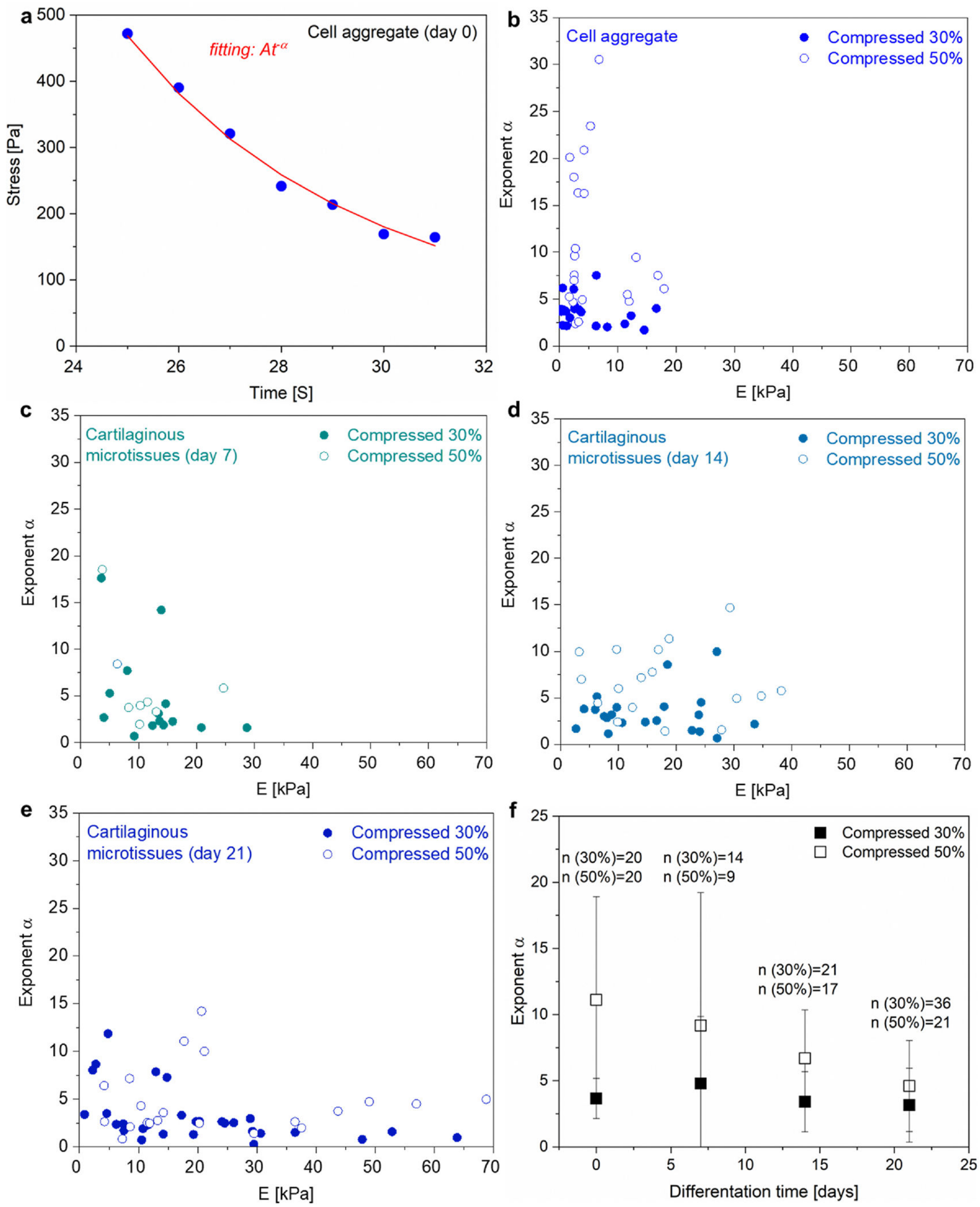

Fig. 8.

Microtissue maturation affects stress dissipation. (a) Example fit of stress dissipation to a power law function for a cell aggregate compressed at 50 %. (b–f) Values of the exponent α for peak compressions of 30 % and 50 % as a function of the Young's modulus of the tissue, corresponding to (b) cell aggregates (day 0), ($n_{30\ \%} = 20, n_{50\ \%} = 20$), (c) cartilaginous microtissues day 7 ($n_{30\ \%} = 14, n_{50\ \%} = 9$), (d) cartilaginous microtissues day 14, ($n_{30\ \%} = 21, n_{50\ \%} = 17$) and (e) cartilaginous microtissues day 21, ($n_{30\ \%} = 36, n_{50\ \%} = 21$). (f)

Average values of α in function of differentiation time, with 0 days corresponding to cell aggregates.

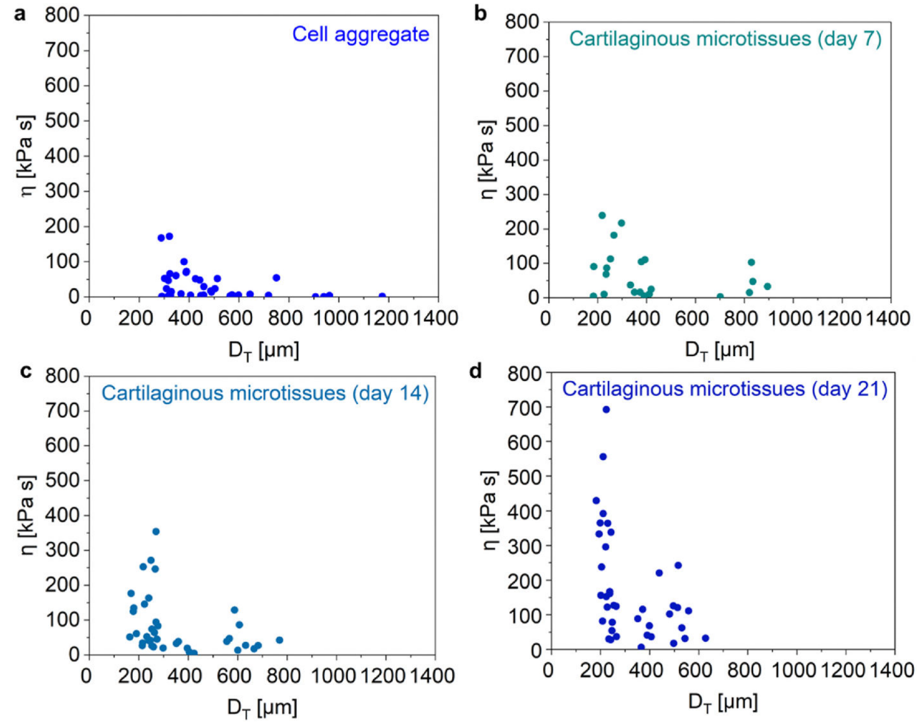

Fig. 9.

Size dependent and increasing viscosity with differentiation time. The viscosity of the tissues in function of the transverse diameter for (a) cell aggregate ($n = 37, p < 0.001$), (a) cartilaginous microtissues day 7 ($n = 22, p < 0.001$), (c) cartilaginous microtissues day 14 ($n = 38, p < 0.001$) and (d) cartilaginous microtissues day 21 ($n = 39, p < 0.001$).

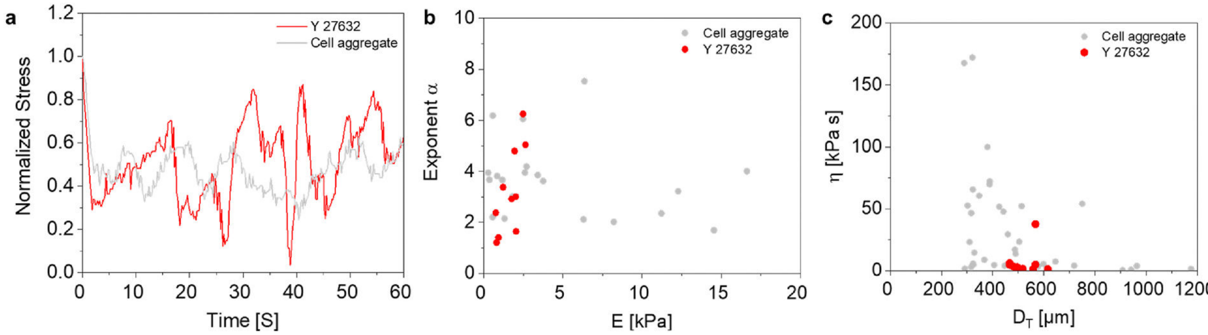

Fig. 10.

Oscillation response depends on actomyosin contractility. (a) Stress relaxation response of cell aggregate treated with 10 μM Y27632 and cell aggregate compressed at 30 %. The exponent α in (b) and the viscosity in (c) for Y27632 treated (n = 10) and non-treated cell aggregates.

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Table 1

Evaluation of the Young’s modulus of the bulk tissue and the thickness of the “tensile” ring based on the fittings of equation (1) to the experimental data σ-E (σ denotes here the IST). In the last column the thickness of the ring is estimated based on the high-resolution images presented in Fig. 4 and Fig. S1.

	E_c [kPa]	*I* [μm]	*I* [μm] (exp.)
Cell aggregate	2.62	15.38	20–40
Cartilaginous microtissues day 7	3.93	12.31	8–35
Cartilaginous microtissues day 14	4.65	8.41	4.7–8
Cartilaginous microtissues day 21	3.93	10.78	2.9–8.3